INCH-POUND

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DETAIL SPECIFICATION

ADHESIVE FOR PLASTIC VIBRATION DAMPING TILES

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a two-part epoxy adhesive for bonding plastic vibration damping tiles to metal structures.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-PRF-23653	-	Plastic Tiles, Vibration Damping
MIL-DTL-24441	-	Paint, Epoxy-Polyamide, General Specification for

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9510-AB-ATM-010	-	Nuclear Powered Submarine Atmosphere Control Manual; Volume 1

(Copies of the chapter titled “Material Control Program” are available by email request to CommandStandards@navy.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D471 - Standard Test Method for Rubber Property – Effect of Liquids

ASTM D2240 - Standard Test Method for Rubber Property – Durometer Hardness

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The adhesive furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Material and form. The adhesive shall be supplied in the form of a two-part system. One part shall contain the epoxy resin and the other part shall contain the hardener (see 6.2). When the two parts of the adhesive are combined in the weight proportions recommended by the manufacturer, the uncured and cured mixture shall meet all applicable requirements of this specification. The adhesive shall not contain solvents of any kind.

3.2.1 Toxicity. When evaluated in accordance with 4.5, the cured adhesive shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.5 and 6.4).

3.2.2 Off-gassing. The cured adhesive shall be evaluated for off-gassing in accordance with the requirements of 4.6. Based on the circumstances of use and the chemical nature of the adhesive, the Navy will determine whether off-gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the cured adhesive shall be assigned to either the “Permitted” or “Limited” category (see 4.6 and 6.5).

3.3 Color. The colors of the two components shall be contrasting when examined in accordance with 4.4.2. Thorough blending of the components shall produce a mixture having homogenous color.

3.4 Viscosity and pot life. The adhesive, after mixing as specified in 3.2 and 4.4.3.1, shall have an initial viscosity of not more than 60 and a pot life viscosity of not more than 75 when tested as specified in 4.4.3.

3.5 Hardness. The hardness of the adhesive shall be 70 minimum at 73.5 °F and 55 minimum at 140 °F when tested as specified in 4.4.4.

3.6 Resistance to accelerated aging. The hardness of the adhesive, after accelerated aging as specified in 4.4.5, shall be 90 maximum and shall not show a change of more than 10 points from its value at 73.5 °F as determined in 3.5.

3.7 Resistance to fuel. The hardness of the adhesive after immersion in the standard test fuel as specified in 4.4.6 shall be 40 minimum.

3.8 Resistance to water. The hardness of the adhesive after immersion in water as specified in 4.4.7 shall be 40 minimum.

3.9 Vertical slippage. The adhesive shall not allow a MIL-PRF-23653 class 1 damping tile weighing 4.5±0.2 lb/ft^2 to slip more than ⅛ inch or to separate from a steel plate when tested as specified in 4.4.8.

3.10 Overhead sag. The adhesive shall not allow a MIL-PRF-23653 class 1 damping tile weighing 4.5±0.2 lb/ft^2 to buckle, sag, or separate from a steel plate when tested as specified in 4.4.9.

3.11 Resistance to shock (high impact). When tested as specified in 4.4.10, there shall be no bond failure between the adhesive and the steel plates or between the adhesive and MIL-PRF-23653 class 1 damping tiles weighing 4.5±0.2 lb/ft^2.

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3.12 Adhesion to painted steel. When tested as specified in 4.4.11, the strength of the adhesive bond shall be not less than 500 psi initially, after accelerated aging, after water immersion, and after immersion in the standard test fuel.

3.13 Adhesion to damping tile. When tested as specified in 4.4.12, the initial strength, strength after accelerated aging, and strength after water immersion of the bond between MIL-PRF-23653 class 1 damping tile and the adhesive shall be not less than 80 psi.

3.14 Storage life. The adhesive shall meet the requirements of this specification after storage for 24 months under the conditions specified in 4.4.13. Furthermore, after storage for 24 months, there shall be no separation of ingredients in either component as determined by visual examination of the contents in each container.

3.15 Identification markings. Unless otherwise specified (see 6.2), each container of the adhesive system shall be marked to provide the following information:

a. Specification number.

b. National stock number.

c. Name of manufacturer.

d. Manufacturer's identification number including designation of component, such as part A or part B.

e. Contract or order number.

f. Date of manufacture.

g. Volume or weight of adhesive component in container.

h. Instruction for preparation of the adhesive including correct proportioning by weight to be used with other part of the adhesive.

i. Recommended storage conditions and limitations on working life.

j. Precautionary markings as follows:

(1) CAUTION: Do not mix resin and hardener together until ready for use.

(2) WARNING: The material in this container may cause skin irritation on some individuals. Avoid skin contact by wearing protective gloves.

(3) Any other handling precautions considered necessary.

(4) Instructions for cleaning equipment used for adhesive application.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspections shall be conducted at a laboratory acceptable to the Naval Sea Systems Command. Government representatives may witness qualification inspections. Qualification inspection shall consist of the tests specified in 4.4, 4.5, and 4.6 (see table I).

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TABLE I. Qualification and conformance inspections.

Characteristic	Qualification	Conformance	Requirement	Verification
Toxicity	X		3.2.1	4.5
Off-gassing	X		3.2.2	4.6
Color	X		3.3	4.4.2
Viscosity and pot life	X	X	3.4	4.4.3
Hardness	X		3.5	4.4.4
Hardness (initial indentation only)		X	3.5	4.4.4
Resistance to accelerated aging	X		3.6	4.4.5
Resistance to fuel	X		3.7	4.4.6
Resistance to water	X		3.8	4.4.7
Vertical slippage	X	X	3.9	4.4.8
Overhead sag	X		3.10	4.4.9
Resistance to shock (high impact)	X		3.11	4.4.10
Adhesion to painted steel	X	X	3.12	4.4.11
Adhesion to damping tile	X		3.13	4.4.12
Storage life	X		3.14	4.4.13
Identification markings	X		3.15	4.4.14

4.3 Conformance inspection. Conformance inspection shall consist of the tests specified in 4.4.3, 4.4.4, 4.4.8, and 4.4.11 (see table I). Government representatives may witness conformance inspections.

4.3.1 Sampling.

4.3.1.1 Identification of lot. For purposes of conformance inspection, a lot shall consist of all adhesive from one production batch offered for delivery at one time.

4.3.1.2 Sampling for conformance inspection. Random samples of the epoxy resin component and the hardener component shall be taken from each lot of filled containers in sufficient quantity to conduct the tests specified in 4.3.

4.3.2 Rejection. The lot shall be rejected if the samples of epoxy resin plus hardener selected in accordance with 4.3.1.2 are found not to conform to any of the requirements of 3.4, 3.5, 3.9, and 3.12.

4.4 Test methods. Unless otherwise indicated in the particular test procedure, tests shall not be conducted prior to conditioning the test specimens in air for a period of 4 hours at 80±9 °F. Sample preparation may be undertaken without regard to this time interval.

4.4.1 Testing conditions. Unless otherwise specified, testing shall be conducted at 80±9 °F.

4.4.2 Color. By visual examination, the colors of the two parts of the adhesive shall be contrasting. Completeness of mixing of the two parts shall be evidenced by disappearance of color streaks and the formation of a homogenous color different than either part of the adhesive.

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4.4.3 Viscosity and pot life. The viscosity of the adhesive shall be determined with a Brookfield viscometer, or equal as approved by NAVSEA, configured for the Brookfield HB torque range. The Brookfield HB series employs a spring torque of 14,374.0 dyne-centimeters (1.4374 millinewton-meters). The viscometer shall be mounted in a Brookfield Helipath Stand or equal as approved by NAVSEA. Measurements shall be made at 73.5±2 °F at 1 revolution per minute with a T-C spindle or equal as approved by NAVSEA.

4.4.3.1 Blending. The two components of the adhesive, in the proportions recommended by the manufacturer, shall be blended together until the mixture is of homogenous color. The time required to do this shall be noted and the blending operation continued for an equal amount of time. At the completion of blending operations, the adhesive shall be immediately transferred to an open-top, cylindrical container. The container shall be at least 2 inches in diameter and the adhesive in the container shall have a depth of at least 3 inches. Care shall be taken to minimize occlusion of air in the adhesive when transferring.

4.4.3.2 Initial viscosity measurement. The supplementary weight furnished with the viscometer shall be attached to the close coupling assembly and the T-C spindle clamped in the chuck. The viscometer shall then be lowered until the cross piece is covered by about ¼ inch of adhesive. Start the viscometer and let the spindle make two complete revolutions before starting the Helipath Stand. Record the next two readings to the nearest whole number after each of the next two complete revolutions of the spindle. The two readings shall be averaged, and the result reported as the initial viscosity. The initial viscosity measurements shall be completed within 30 minutes after the start of blending operations.

4.4.3.3 Final viscosity measurement. Immediately after completing initial viscosity measurements, the adhesive shall be removed from the cylindrical container and spread out on a metal plate to a maximum depth of ¼ inch. After 1 hour and 25 minutes following the start of blending operations, the adhesive shall be removed from the metal plate, placed again into the cylindrical container, and the viscosity redetermined as described in 4.4.3.2. The viscosity determined shall be termed the pot life viscosity.

4.4.4 Hardness. Adhesive hardness shall be measured with a durometer in accordance with ASTM D2240, type D.

4.4.4.1 Specimen preparation. The hardness shall be measured on specimens approximately 2 inches in diameter by 0.5±0.1 inch thick and having smooth, parallel faces. The specimens shall be prepared by casting the mixed adhesive in an open mold. The adhesive shall be mixed as specified in 4.4.3.1. While the adhesive is still in a semi-fluid state, the mold and contents shall be placed in a chamber and subjected to 1,000 to 1,500 psig nitrogen gas pressure at room temperature for 16 to 24 hours. The specimens shall then be removed from the mold and placed in an air-circulating oven for 5 days at 120±2 °F. For acceptance testing, the stabilizing operation may be omitted if desired; however, the test specimens shall still meet the requirements for initial hardness. The specimens shall then be machined to the required dimensions. Three test specimens shall be prepared.

4.4.4.2 Test procedure. The specimens shall be conditioned at the testing temperature for a minimum of 2 hours immediately before the start of test operations. The indenter of the durometer shall be pressed firmly into one flat side of the specimen and the durometer indication shall be noted 3 seconds after initial indentation. The base of the durometer shall press against a smooth machined surface of the specimen and the indenter shall be not closer than ¼ inch to the edge of the specimen. Three readings shall be taken, each at least ¼ inch apart on each of the three specimens, and the average hardness shall be reported for each specimen.

4.4.5 Resistance to accelerated aging. Resistance to accelerated aging shall be determined by the change in hardness produced by conditioning the hardness specimens in an air-circulating oven for 2 weeks at 158±2 °F. Three test specimens prepared as described in 4.4.4.1 shall be used. At the conclusion of the aging period, the specimens shall be removed from the oven and allowed to cool at 73.5±2 °F for at least 2 hours. Hardness measurements shall then be made as specified in 4.4.4.2.

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4.4.6 Resistance to fuel. Resistance to fuel shall be determined by the change in hardness resulting from immersion of the specimens in Reference Fuel B specified in ASTM D471. This fuel is a 70/30 mixture by volume of iso-octane with toluene. Three specimens prepared as specified in 4.4.4.1 shall be used in this test. Each specimen shall be placed in a separate container and the specimen covered with a volume of fuel equivalent to 12 to 20 times the volume of the test specimen. The immersed specimens shall be kept at 73.5±2 °F for 2 weeks. The specimens shall then be removed, one at a time, from the fuel, blotted dry, and their hardness determined within 1 minute after removal from the fuel as specified in 4.4.4.2.

4.4.7 Resistance to water. Resistance to water shall be determined by the change in hardness resulting from immersion of the specimens in distilled water. Three specimens prepared as specified in 4.4.4.1 shall be used in this test. Each specimen shall be placed in a separate container and the specimen covered with a volume of water equivalent to 12 to 20 times the volume of the test specimen. The immersed specimens shall be kept at 120±2 °F for a period of 2 weeks. At the conclusion of this conditioning period, the water shall be poured off and a fresh portion of cool distilled water poured over the specimens. The specimens shall be allowed to cool at 73.5±2 °F in the water for at least 2 hours but not more than 4 hours. They shall then be removed, one at a time, blotted dry, and their hardness determined within 1 minute after removal from the water as specified in 4.4.4.2.

4.4.8 Vertical slippage. Vertical slippage of a MIL-PRF-23653 class 1 damping tile weighing 4.5±0.2 lb/ft^2 applied to a simulated vertical installation shall be measured at 73.5±2 °F as follows:

a. A flat steel plate ⅜ by 16 by 16 inches in size shall be painted with a type III or type IV paint system in accordance with MIL-DTL-24441. The plate shall be degreased and blasted prior to painting. The plate shall not show any signs of curvature after blasting. The paint on the plate shall be allowed to cure 1 week minimum, 4 months maximum, prior to use of the plate.

b. The adhesive, mixed as specified in 4.4.3.1, shall be applied to one side of the tile and to the central foot-square area of the painted steel plate with a serrated tool having equilateral triangular grooves in the spreading edge. The grooves shall intersect each other and shall be 3/16 inch deep. The tile shall be placed against the coated steel plate so that the rows of adhesive formed by the serrated tool on the tile and on the plate are parallel and run from the edge of the tile designated top to the edge of the tile designated bottom. The tile shall then be pressed firmly against the plate with a hand roller, thus forcing all excess adhesive to be extruded from beneath the tile. This excess adhesive shall be removed and discarded. A horizontal line shall be scribed across the plate at the edge of the tile designated bottom.

c. The assembly shall be secured in a vertical position with the edge of the tile designated bottom facing downward. The installation of the tile and the securing of the assembly shall be accomplished within 30 minutes after mixing the adhesive.

d. The amount of vertical slippage, buckling of the tile, or separation of the tile from the surface shall be noted and recorded a minimum of 12 hours after placement of the assembly in a vertical position.

4.4.9 Overhead sag. Overhead sag of a MIL-PRF-23653 class 1 damping tile weighing 4.5±0.2 lb/ft^2 applied in a simulated overhead installation shall be measured at 73.5±2 °F as follows:

a. An assembly of a steel plate and damping tile shall be prepared exactly as specified in 4.4.8.a and 4.4.8.b, except that no reference line need be drawn.

b. The assembly shall be placed in a horizontal position with the steel plate supported at the edges and the side with installed tile facing downward. The installation of the tile on the plate and the positioning of the assembly shall be accomplished within 30 minutes after mixing the adhesive.

c. The amount of sag, buckling of the tile, or separation of the tile from the surface shall be noted and recorded a minimum of 12 hours after placement of the assembly in a horizontal position.

4.4.10 Resistance to shock (high impact). The resistance of the adhesive to mechanical shock shall be determined on the lightweight shock machine specified in MIL-DTL-901 using test specimens consisting of damping tiles bonded to painted steel plates.

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4.4.10.1 Specimen preparation.

4.4.10.1.1 Substrate plate preparation. Five steel plates ⅜ by 16 by 16 inches in size shall be painted with a type III or type IV paint system in accordance with MIL-DTL-24441. The plates shall be degreased and blasted prior to painting operations. The plates shall not show any signs of curvature after blasting. The paint on the plates shall be allowed to cure for 1 week minimum and 4 months maximum before installing tiles on the painted surfaces.

4.4.10.1.2 Bonding damping tile. One MIL-PRF-23653 class 1 damping tile weighing 4.5±0.2 lb/ft^2 shall be bonded to the central portion of each painted plate in the manner specified in 4.4.8.b, except that no reference line need be scribed. The assemblies may be placed in a plastic bag and the air evacuated in order to obtain a uniform pressure across the face of the tile until the adhesive sets. The assemblies shall be kept at ambient room temperature for 16 to 24 hours and then they shall be placed in an air-circulating oven at 120±2 °F for 5 days.

4.4.10.2 Testing for shock resistance.

4.4.10.2.1 Specimen cooling and mounting. The assemblies shall be mounted in a vertical position on the central portion of the anvil plate of the shock machine. Mounting fixtures, such as the 4A plate, shall not be used. The assembly shall be secured with four ⅞-inch diameter bolts, one located in each of the four corners. The assembly shall be held 1 inch away from the anvil plate by means of spacers 2 inches in diameter placed around each securing bolt. Prior to mounting the assembly in the shock machine, it shall be cooled to 32 to 35 °F by placing in a mixture of crushed ice and water for at least 2 hours. Shock testing of the assembly shall be accomplished within 10 minutes after removal from the cooling mixture.

4.4.10.2.2 Impact sequence. Each assembly shall be subjected to the following sequence of blows from the falling weight:

a. Two 5-foot back blows.

b. Two 5-foot top blows – retighten all hold-down bolts.

c. Two 5-foot top blows.

d. Two 5-foot back blows.

Examine the assembly for bond failure after each blow from the falling weight.

4.4.11 Adhesion to painted steel. Adhesion to painted steel shall be determined in tension as described in the following paragraphs.

4.4.11.1 Specimen preparation. Determination of adhesion to painted steel requires 12 test specimens. Each test specimen shall consist of two flat steel disks, 1.597 inches in diameter by ⅜ inch in thickness, bonded together with the adhesive. The opposite face of each steel disk shall have a centrally located threaded stud ½ inch in diameter by 1 inch long. One face of each of the metal disks shall be degreased, blasted, and painted with a type III or type IV paint system in accordance with MIL-DTL-24441. The paint shall be allowed to cure for 1 week minimum, 4 months maximum, before using the disks. The adhesive shall be mixed as specified in 4.4.3.1 and applied to the painted faces of the metal disks, using a hacksaw blade with 18 teeth per inch as an applicator. The coated sides of the disks shall be pressed together with sufficient pressure to cause the adhesive to extrude from between the faces of the disks. The final adhesive layer shall not exceed 1/32 inch in thickness. Twelve replicate specimens shall be prepared. The assembled specimens shall be kept at ambient room temperature for 16 to 24 hours and then they shall be placed in an air-circulating oven at 120±2 °F for 5 days. This stabilizing cure may be omitted if desired when preparing specimens for the conformance inspections specified in 4.3. A drawing of an assembled test specimen is shown on figure 1.

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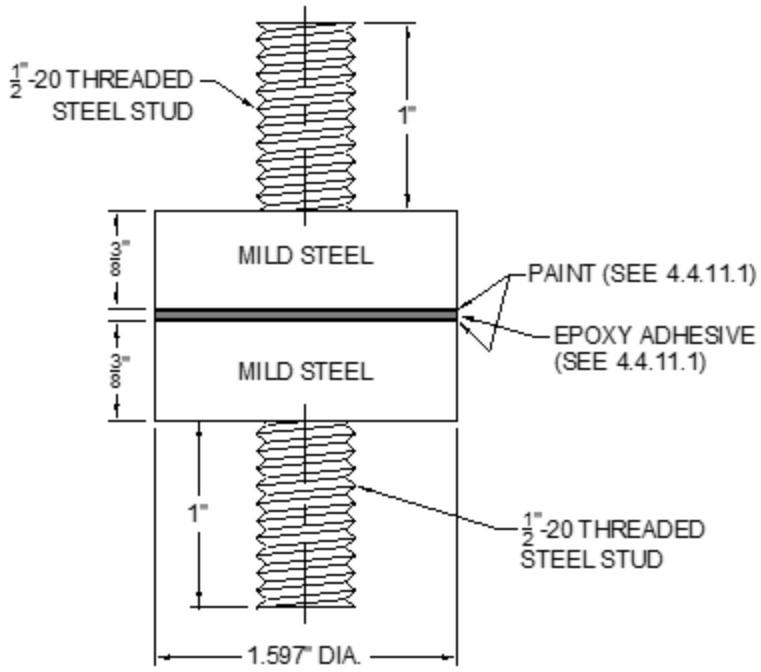


FIGURE 1. Test specimen for measuring adhesion to painted steel.

4.4.11.2 Conditioning. After completion of curing operations, three specimens shall be exposed to a temperature of 158±2 °F in an air-circulating oven for 2 weeks, three specimens shall be immersed in 12 to 20 times their volume of distilled water at 120±2 °F temperature for 2 weeks, and three specimens shall be immersed in 12 to 20 times their volume of Reference Fuel B of ASTM D471 (see 4.4.6) for 2 weeks at 73.5±2 °F. The three remaining specimens shall be used for measuring initial bond strength.

4.4.11.2.1 Cooling. After conclusion of the conditioning specified in 4.4.11.2, the specimens that were conditioned at 158 °F shall be allowed to cool in air at 73.5±2 °F for 16 to 24 hours before testing for bond strength. The specimens that were conditioned in water at 120 °F shall be placed in a fresh portion of cool water, conditioned in water at 73.5±2 °F for a minimum of 2 hours but not more than 4 hours, and tested for bond strength immediately on removal from the water. The specimens that were immersed in Reference Fuel B of ASTM D471 at 73.5 °F shall be tested for bond strength immediately after the conclusion of the conditioning period specified in 4.4.11.2.

4.4.11.3 Testing for bond strength. Bond strength shall be determined at 73.5±2 °F. The test specimens shall be placed individually in a tension-testing machine and the force required to separate the two disks measured. The moving head of the machine shall travel at a rate of 0.02 inch per minute. The results obtained on each group of three specimens shall be averaged and divided by two to give the results in terms of pounds per square inch.

4.4.12 Adhesion to damping tile. The bond between the adhesive and the damping tile shall be determined as described in the following paragraphs.

4.4.12.1 Specimen preparation. Nine disks, each 4 inches in diameter, shall be cut from a tile in accordance with class 1 of MIL-PRF-23653. One face of each disk shall be bonded with properly mixed adhesive to a steel disk ½ inch thick by 4 inches in diameter that has been cleaned by blasting and painted as described in 4.4.11.1. The opposite face of the steel disk shall have a centrally located threaded stud ½ inch in diameter by 1 inch long. A steel disk, 1.597 inches in diameter by ⅜ inch thick, with a threaded stud on one face shall be degreased, cleaned by blasting, painted as described in 4.4.11.1, and then bonded with the adhesive in a central position to the opposite face of the tile disk. The adhesive shall be mixed as specified in 4.4.3.1 and then applied to the steel disks only in the same manner as 4.4.11.1. Nine replicate specimens shall be prepared. The assembled specimens shall be kept at ambient room temperature for 16 to 24 hours and then they shall be placed in an air-circulating oven at 120 °F for 5 days. A drawing of an assembled specimen is shown on figure 2.

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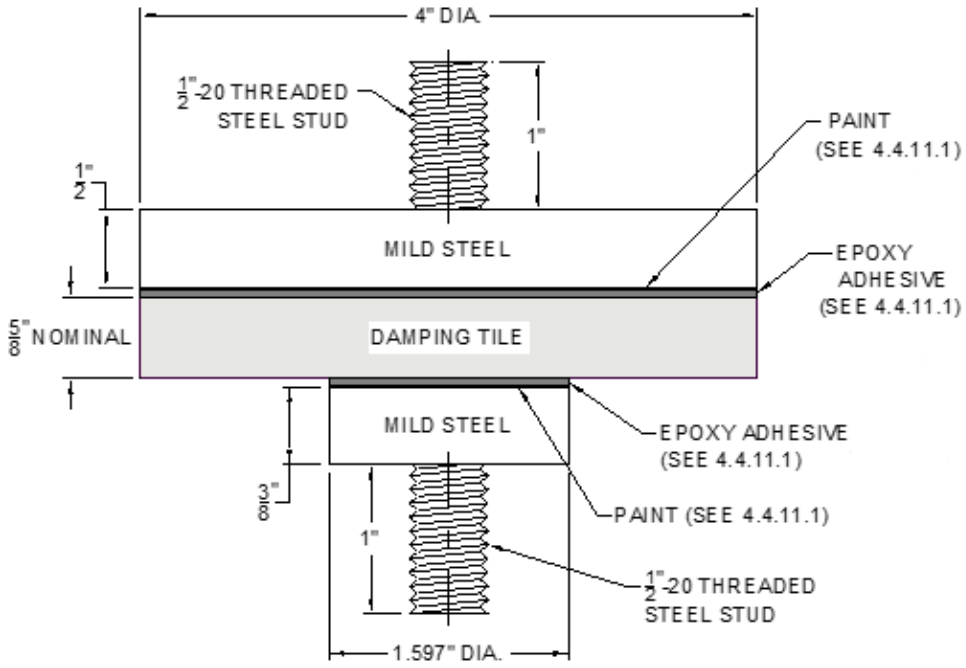


FIGURE 2. Test specimen for measuring adhesion to plastic damping tile.

4.4.12.2 Conditioning. After completion of curing operations, three specimens shall be conditioned in an air-circulating oven at 158±2 °F for 2 weeks and three specimens shall be conditioned in water at 120±2 °F for 2 weeks in the same manner as specified in 4.4.11.2.

4.4.12.3 Testing for bond strength. Bond strength shall be determined at 73.5±2 °F in the same manner as specified in 4.4.11.3 except that the rate of separation of the disk from the specimen shall be 1 inch per minute.

4.4.13 Storage life. The adhesive shall be tested after being stored in its original unopened container for 24 months at the temperature specified in 4.4.1 and shall meet the requirements of the conformance inspections of table I.

4.4.14 Identification markings. Verify by visual examination that identification markings on each container of the adhesive system conform to 3.15.

4.5 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.2.1 as required by the qualifying activity. The Navy and Marine Corps Force Health Protection Command (NMCFHPC) will evaluate the adhesive using data provided by the manufacturer/distributor to the NMCFHPC (see 3.2.1 and 6.4).

4.6 Off-gassing. The adhesive shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010 chapter titled “Material Control Program” (see 3.2.2 and 6.5). If the Navy determines that off-gas testing is required, testing shall be conducted at a NAVSEA approved test facility (see 3.2.2). The Navy will review the off-gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off-gas testing (see 3.2.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity (see 6.2).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The adhesive covered by this specification is intended for use aboard Navy ships to bond vibration damping material including MIL-PRF-23653, MIL-P-22581, and MIL-DTL-24487 tiles, and other noise and vibration control treatments, to ship structures. The properties of this adhesive are integral to the performance of those treatments and ensure Navy ships meet operational and lifecycle requirements. Commercial off-the-shelf adhesive will not support Navy needs unless the adhesive complies with the requirements of this specification. Current stock of adhesive may be used until depleted; future requisitions for adhesive should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Material and form (see 3.2).

c. Container markings if other than specified (see 3.15). Note that the following are considered the minimum necessary for the safe, effective, and proper identification, storage, use, and disposal of this adhesive:

(1) Specification number.

(2) National stock number.

(3) Name of manufacturer.

(4) Manufacturer's identification number including designation of component such as part A or part B.

(5) Date of manufacture.

(6) Volume or weight of adhesive component in container.

(7) Instructions for preparation of the adhesive including correct proportioning by weight to be used with other part of the adhesive.

(8) Recommended storage conditions.

(9) Limitations on working life.

(10) Handling precautions considered necessary including safety requirements and not mixing resin and hardener until ready for use.

(11) Instructions for cleaning equipment used for application.

d. Quantity desired, container size, and packaging requirements (see 5.1).

e Coverage, in square feet, provided by each unit of issue, assuming a uniform layer of adhesive 1/32 inch thick.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24456 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Toxicity evaluation. The NMCFHPC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCFHPC to the Government for evaluation. The HHA process is described on the NMCFHPC's website, https://www.med.navy.mil/Navy-and-Marine-Corps-Force-Health-Protection-Command/Environmental-Health/Industrial-Hygiene/Acquisition-Technical-Support/Health-Hazard-Assessments-HHA/.

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6.5 Material certification. Materials to be installed in submarines are to be controlled to prevent off-gassing, which contaminates the submarine's atmosphere and can result in health hazards to personnel or deleterious effects on machinery. These controls are administered through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, S9510-AB-ATM-010 chapter titled "Material Control Program." Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by Navy activities or contractors, and a request for certification is submitted to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. The certification request is accompanied by detailed information, including descriptions of the material, method of application, usage, and storage. A chemical analysis is conducted, which can be accomplished through off-gas testing. If off-gas testing is required, it must be conducted in a Government approved laboratory. Information pertaining to this test requirement may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. Based on the chemical analysis results, a usage category is assigned to the material defining whether, and to what extent, the material may be used on submarines.

6.6 Major defects. Table II lists major defects recommended for lot rejection based on the recommended minimum 5 percent sampling rate (a greater sampling rate may be warranted for small orders).

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TABLE II. Major defects in adhesive containers, packing, and shipping containers.

Adhesive containers
Evidence of leakage
Not type specified
Warning or caution marking on labels not provided
Capacity not as specified
Contents below the required or indicated fill
Metal thickness not in accordance with type of container and product contained
Gasket (if required) missing, not seated, or damaged
Internal coating on metal not as specified
Rust or corrosion on exterior
Packing of containers
Not conforming to carriers' rules and regulations
Not the level specified
Separators, buffer pads, or cell spacers not as specified
Shipping containers
Not type specified
Damaged
Not filled
Reinforcing tape (when required) not type specified or correctly applied
Closing or strapping not as specified
Palletized containers (when specified) not correctly assembled

6.7 Minor defects. Table III lists minor defects recommended for consideration of lot rejection based on the recommended minimum 5 percent sampling rate (a greater sampling rate may be warranted for small orders).

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TABLE III. Minor defects in adhesive containers, packing, and shipping containers.

Defects
Adhesive containers
Not of same type and capacity
Dented, scratched, or dirty
Marking incomplete, incorrect, or illegible
Bail or handle (when specified) missing or damaged
Bail grip (when specified) damaged
Packing of containers
Closures not as specified
Handles or bails not protected
Arrangement of product containers not as specified
Marking incomplete, incorrect, or illegible
Shipping containers
Marking incorrect, incomplete, or illegible

6.8 NAVSEA contact information. Any questions concerning references herein to "Naval Sea Systems Command" or to "NAVSEA" should be referred to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil.

6.9 Subject term (key word) listing.

Bonding
Constrained-layer
Free-layer
Gap filler
Grout
Plating, steel
Restrained-layer
Structures, metal

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 8040-2024-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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